

Practice Exam 4 (A)

(See also the fluid pressure problems on the Koen practice exam.)

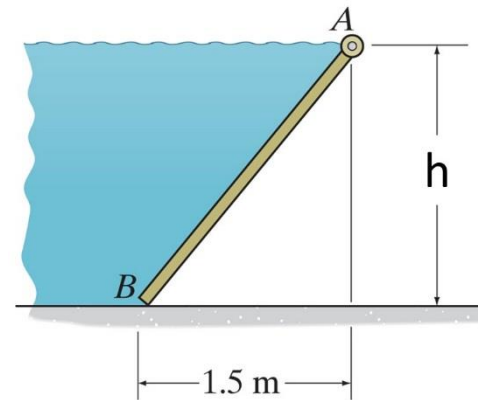
1 What is the simplest shape that represents the total loading distribution on plate AB ?

- A. trapezoid
- B. triangle
- C. square
- D. rectangle
- E. parabolic curve

2 Where would the center of pressure for the hydrostatic force acting on gate AB be located along AB measured from A ?

$h = 2 \text{ m}$

- A. 1.67 m
- B. 0.83 m
- C. 1.12 m
- D. 2.24 m
- E. 0.75 m



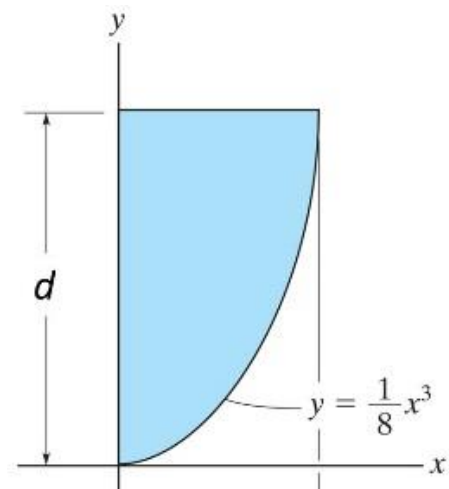
Section 2a Moment of Inertia

3. Determine the moment of inertia of the shaded area about the x axis. $d = 8 \text{ m}$

- A. 538 m^4
- B. 614 m^4
- C. 702 m^4
- D. 910 m^4
- E. 869 m^4

4. Which initial expression would be integrated to determine the moment of inertia with respect to the y axis when using a horizontal differential element and single integration for the shaded area?

- A. $(x^2 dy)$
- B. $(y^2 dx)$
- C. $(.125 x^3 dx)$
- D. $(x^3 / 12 + x^3 / 4) dy$
- E. $(x^3 / 12 - x^3 / 4) dy$



The composite shape can be viewed as a rectangle plus a triangle, with a circle subtracted out.

The parallel axis theorem ($I = \bar{I} + Ad^2$) can be used to determine the moment of inertia about the x axis.

5. What is the magnitude of “I” for the circle?

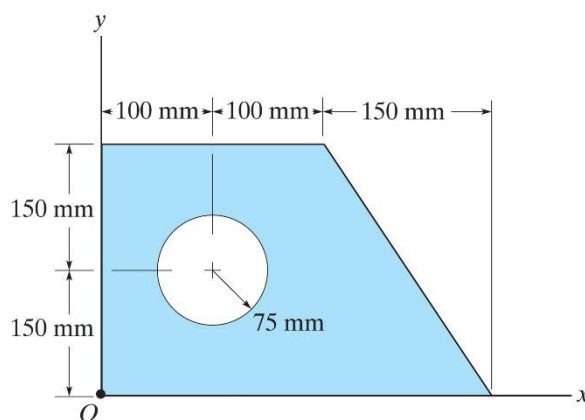
- A. $3.10 (10^8) \text{ mm}^4$
- B. $2.02 (10^8) \text{ mm}^4$
- C. $5.31 (10^8) \text{ mm}^4$
- D. $1.73 (10^8) \text{ mm}^4$
- E. $4.22 (10^8) \text{ mm}^4$

6. What is $\bar{I}$ for the rectangle?

- A. $450 (10^6) \text{ mm}^4$
- B. $200 (10^6) \text{ mm}^4$
- C. $300 (10^6) \text{ mm}^4$
- D. $350 (10^6) \text{ mm}^4$
- E. $275 (10^6) \text{ mm}^4$

7. What is “d” for the triangular area?

- A. 50 mm
- B. 200 mm
- C. 250 mm
- D. 100 mm
- E. 300 mm

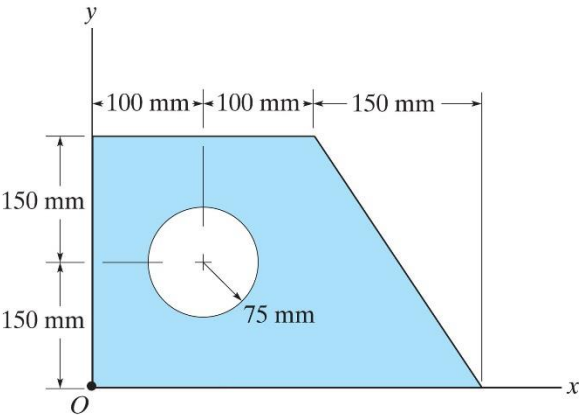


Section 3 – Product of Inertia

POI Concepts:

1. What are units of MOI? POI?
2. How do areas (A) and distance (d) affect MOI? POI?
3. Does the location with respect to the axes affect POI? How?
4. If the area is more in quadrant I or II or III or IV—how will each affect POI?
5. What if an area is symmetric, how does that affect POI?
6. What do values of POI mean? What if POI for a shape (and for a set of axes is zero)?
7. Understand each term of the POI PAT: $I_{xy} = \bar{I}_{x'y'} + Adxdy$
8. What are possible values of $\bar{I}_{x'y'}$? Of dx and/or dy?

9. For the shape at right, what are $\bar{I}_{x',y'}$ and A and dx and dy values for each of the component shapes?



Shape	$\bar{I}_{x',y'}$	A	dx	dy
Rectangle				
Circle				
Triangle				

Mohr's Circle

Question 1

Given I_{AVG} and diameter, find I_{MAX} and I_{MIN} .

Center I_{AVG} =	12.00	in ⁴
Diameter d =	10.00	in ⁴

Question 2

Given I_{AVG} and point B [I_u, I_{uv}]:

Center I_{AVG} =

12.00

in⁴

Coordinates of point B [I_B, I_{BD}]:

I_B =

19.88

in⁴

I_{BD} =

6.16

in⁴

Please determine:

Radius R, I_{MAX} , I_{MIN} , point D [I_D, I_{BD}] (90 deg from point B on area), θ_{p1} , θ_{p2}

The diagram shows a circle with center C. A horizontal line passes through C, and a vertical line is tangent to the circle on the left. Point B is located on the upper right portion of the circle's circumference.